Cortical phase-amplitude coupling is key to the occurrence and treatment of freezing of gait

Paper: Yin et al. (2022). Human walking study.

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Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

Measurement, freezing of gait, and the choice of a clinical control objective. Tests whether a cortical coupling measure tracks freezing and changes with STN stimulation.

How does it answer the general questions?

Measured and manipulated. Motor-cortex ECoG and gait were recorded during walking tasks in 16 patients with Parkinson's disease. Beta–gamma phase-amplitude coupling (PAC) was higher in trials with freezing and related to freezing severity. STN-DBS reduced PAC and alleviated freezing (PDF pp. 1–10).

Usable insight. Freezing still improved with DBS when comparing trials with similar PAC. Therefore, PAC reduction alone does not explain the benefit. A proposed "bandwidth" account combines baseline synchrony, task demands, and processing capacity; it is a conceptual explanation rather than an identified dynamical model (PDF pp. 9–13).

Model relevance. The supported behavioural objective is reducing freezing while preserving walking. PAC is a candidate feedback feature, requiring a nonlinear signal-processing block beyond the notebook's linear measurement equation. The paper supplies no delayed rate-model parameterization or implemented PAC-based adaptive controller.

What papers does it cite that could provide more information?

- Pozzi et al. (2019), *Freezing of gait in Parkinson's disease reflects a sudden derangement of locomotor network dynamics*: network context (ref. 13, PDF p. 14).
- Cole et al. (2017), *Nonsinusoidal beta oscillations reflect cortical pathophysiology in Parkinson's disease*: waveform contributions to PAC (ref. 41, PDF p. 14).